Installing Red Hat Enterprise

Linux Server

Practical 0: Installation of RHEL6.x

There is a version of RH EL Server for almost any hardware platform. That means you can install it on a mainframe computer, a mid-range system, or PC-based server hardware using a 64- or 32-bit architecture.

The ideal installation is on server-grade hardware. However, you don't have to buy actual server hardware if you just want to learn how to work with Red H at Enterprise Linux. Basically, any PC will do as long as it meets the following minimum requirements:

A CPU capable GB of RAM
20GB of available hard disk space
A DVD drive
A network card of handling 64-bit instructions
Apart from these requirements, other requirements relate to KVM virtualization as well. The most important of these is that the CPU on your computer needs virtualization support.
Preinstallation tep : Download ISO
Visit: https://www.redhat.com and download the RHEL 6 ISO.

1. Put the RHEL 6 installation disc in the optical drive of your computer, and boot from the installation disc. If the DVD drive is not in the default boot order on your computer, you'll have to go into the setup and instruct your computer to boot from the optical drive. After booting from the installation DVD successfully, you'll see the Wel- com e to Red Hat

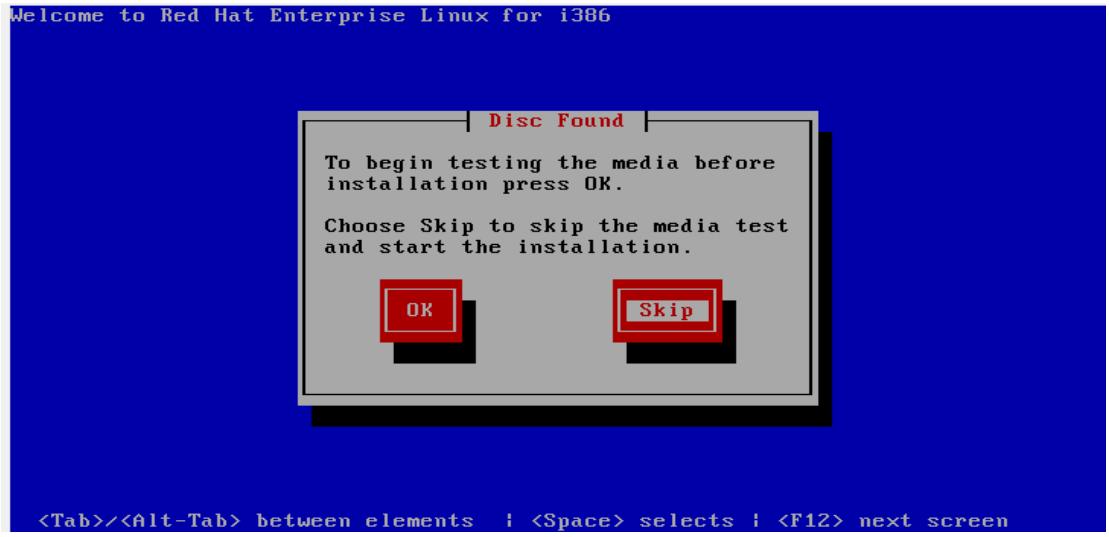
Enterprise Linux screen.



1. From the graphical installation screen, select Install Or Upgrade An Existing System . In case you're experiencing problems with the graphical display, you can choose to install using the basic video driver. However, in most cases that isn't necessary. The other options are for troubleshooting purposes only and w ill be discussed in later chapters in this book.

2. After beginning the installation procedure, a Linux kernel is started, and the hardware is detected. This normally takes about a minute.

3. Once the Linux kernel has been loaded, you will see a non graphical screen that tells you that a disc was found. (Non Graphical enus like the one in the following image are referred to as *ncurses interfaces*. *Ncurses* refers to the programming library that was used to create the interface.)

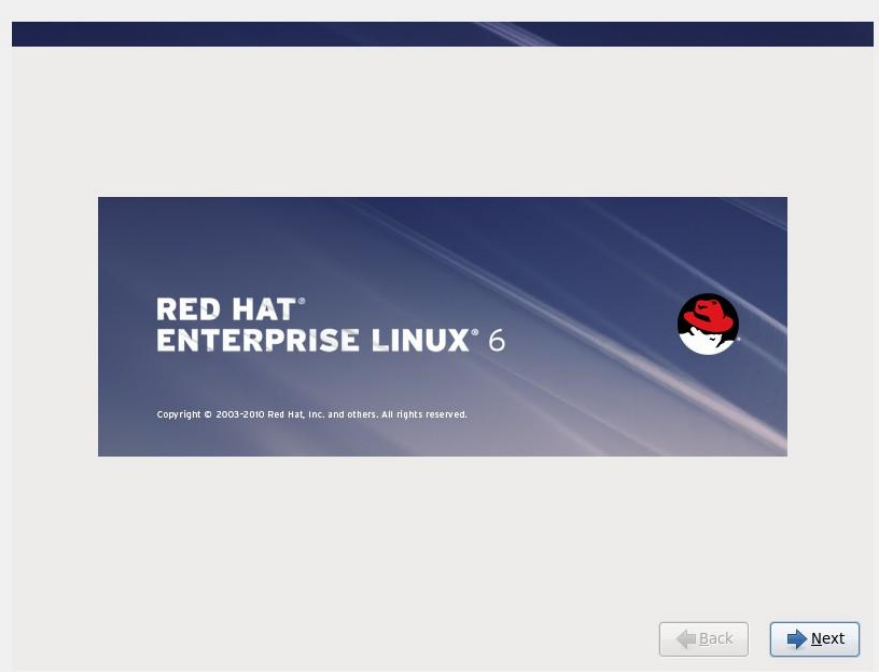


From this screen, you can start a check of the integrity of the installation
media. Don't do this by default; the media check can easily take 10 m inutes or more! Press the Tab key once to navigate to the Skip button, and press Enter to proceed to the next step.

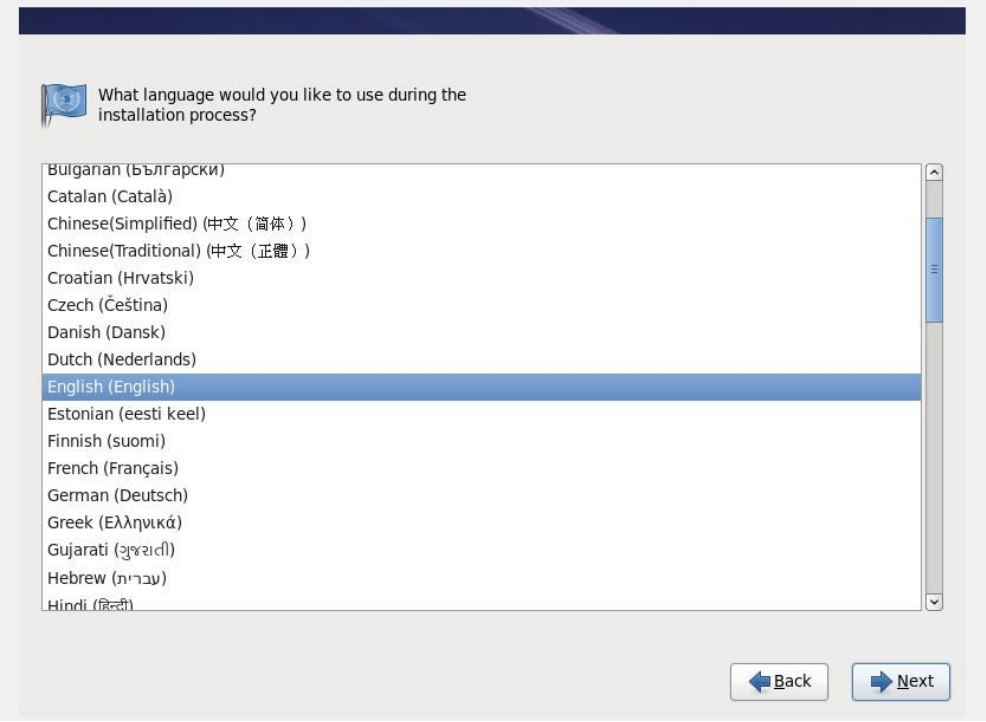
4. If the graphical hardware in your computer is supported, you'll next see a graphical screen with only a Next button on it. Click this button to continue.

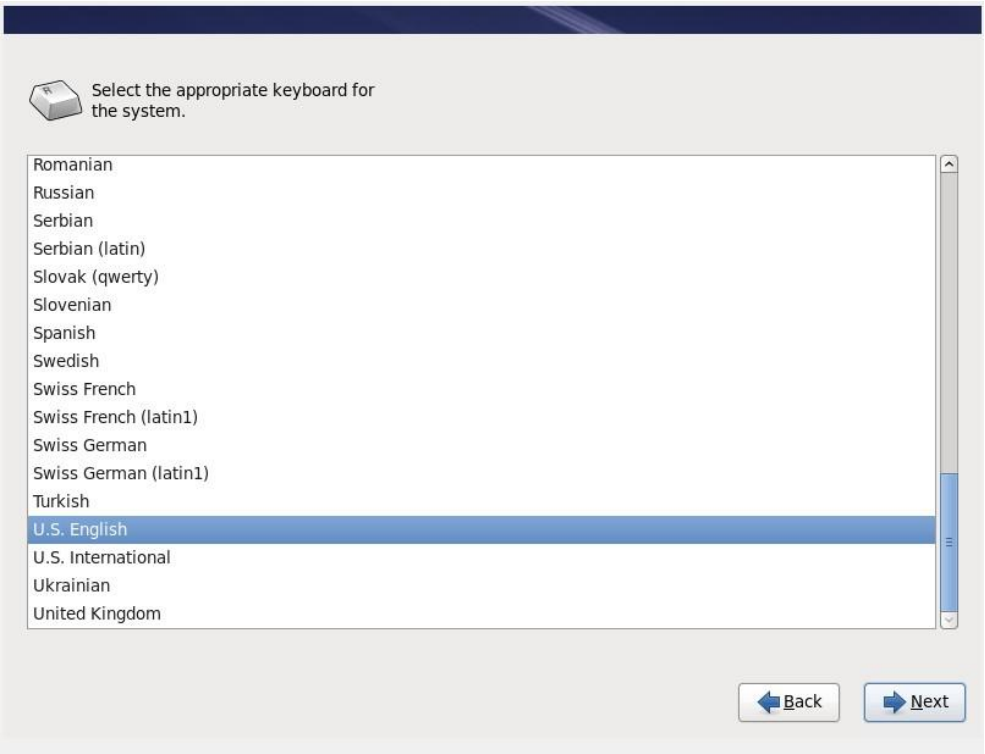
If you don't see the graphical screen at this point, restart the installation pro-cedure by rebooting your computer from the installation disc. From the menu, select Install System With Basic Video Driver.





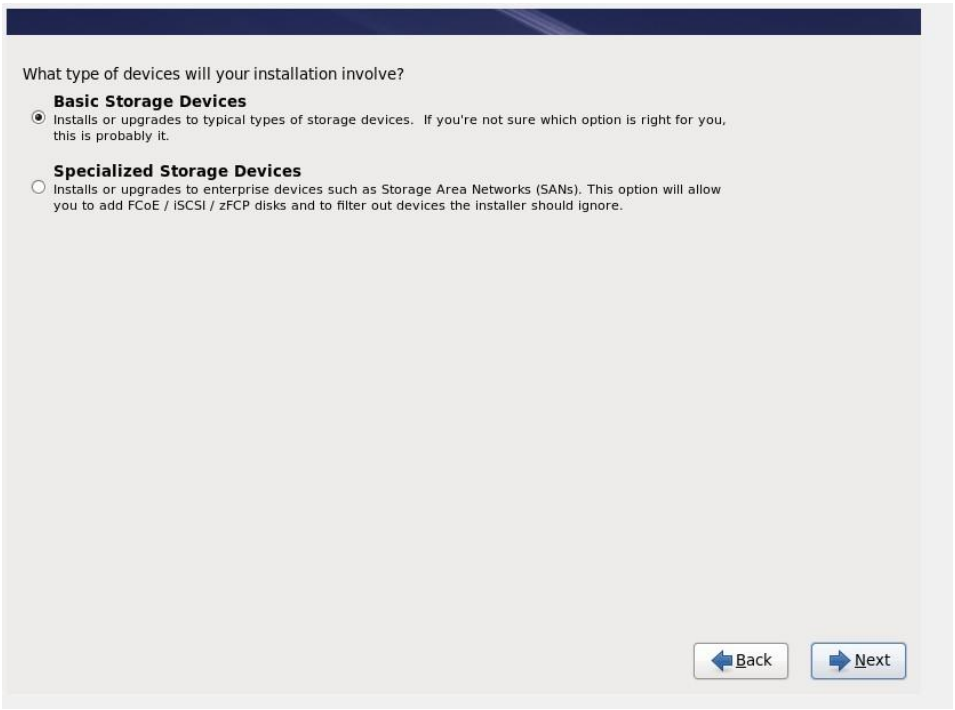
5. On the next screen, you can select the language you want to use during the instal- lation process. This is just the installation language. At the end of the installation, you'll be offered another option to select the language you want to use on your Red Hat server. Many languages are supported; in this book I'm using English.





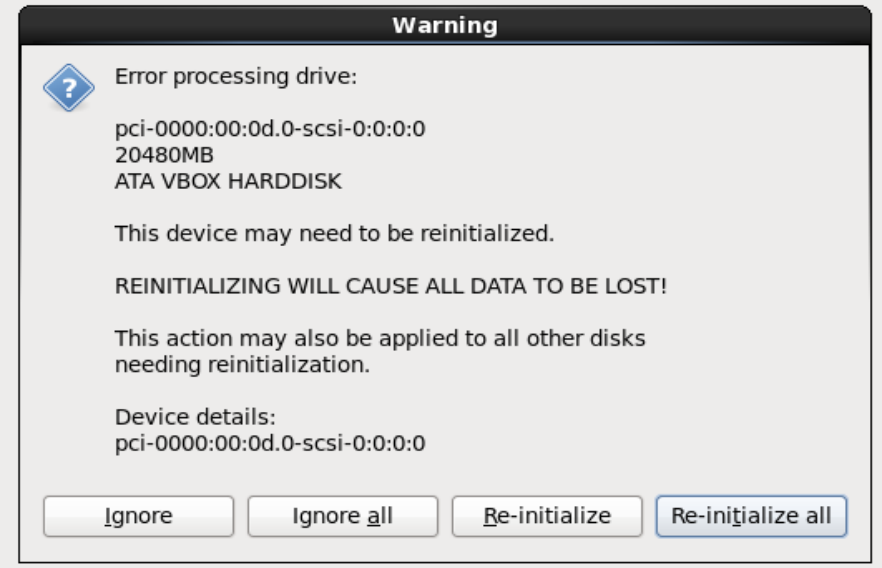
6. After selecting the installation language, on the next screen, select the appropriate keyboard layout, and then click Next to continue.

7. Once you've selected the keyboard layout you want to use, you need to select the storage devices with which you are working. To install on a local hard drive in your computer, select Basic Storage Devices. If you're installing RHEL in an enterprise environment and want to write all files to a SAN device, you should select the Specialized Storage Devices option. If you're unsure about what to do, select Basic Storage Devices and click Next to proceed.

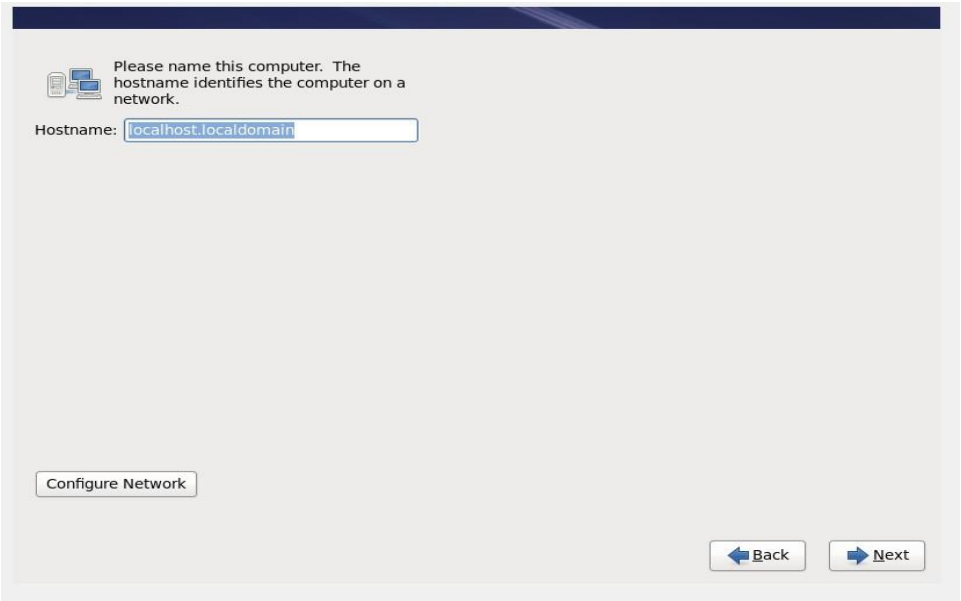


8. After you have selected the storage device to be used, the installation program may issue a warning that the selected device may contain data. This warning is displayed to prevent you from deleting all the data on the selected disk by accident. If you're sure that the installer can use the entire selected hard disk, click Yes,

and discard any data before clicking Next to continue.

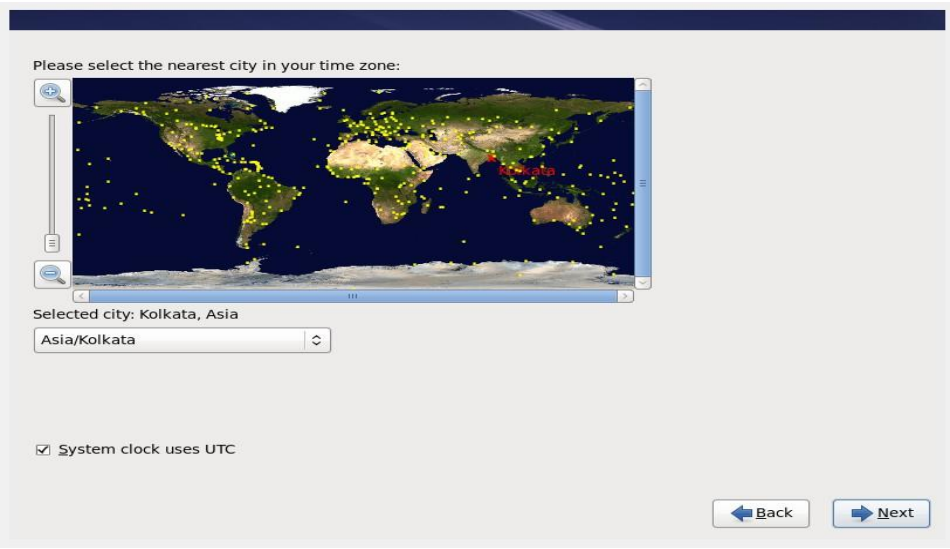


9. On the next screen, you can enter the hostname you want to use on the computer. Also on this screen is the Configure Network button, which you'll use to change the current network settings for the server. Start by entering the hostname you want to use. Typi- cally, this is a fully qualified domain name that includes the DNS suffix. If you don't have a DNS domain in which to install the server, you can use example.com. This name is available for test environments, and it won't be visible to others on the Internet.



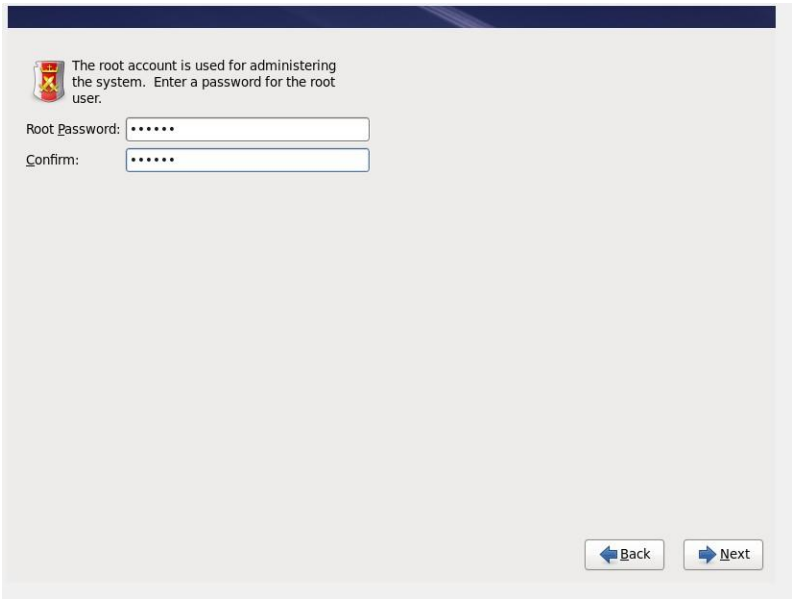
10. Click Next to continue.

11. At this point, you'll configure the time settings for your server. The easiest way to do this is just to click the city nearest to your location on the world map that is dis- played. Alternatively, you can choose the city that is nearest to you from the drop- down list.

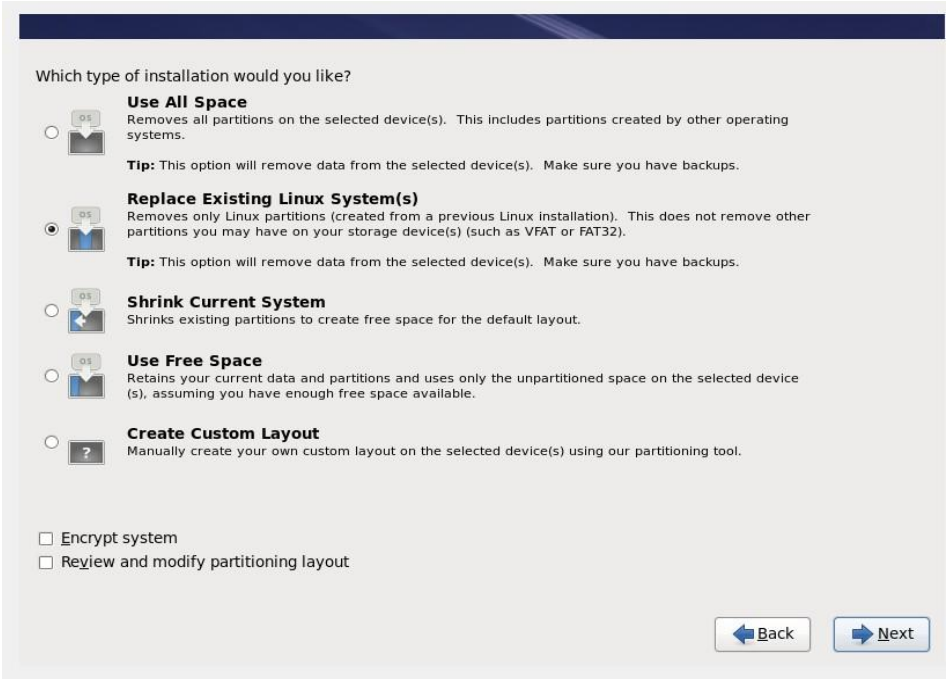


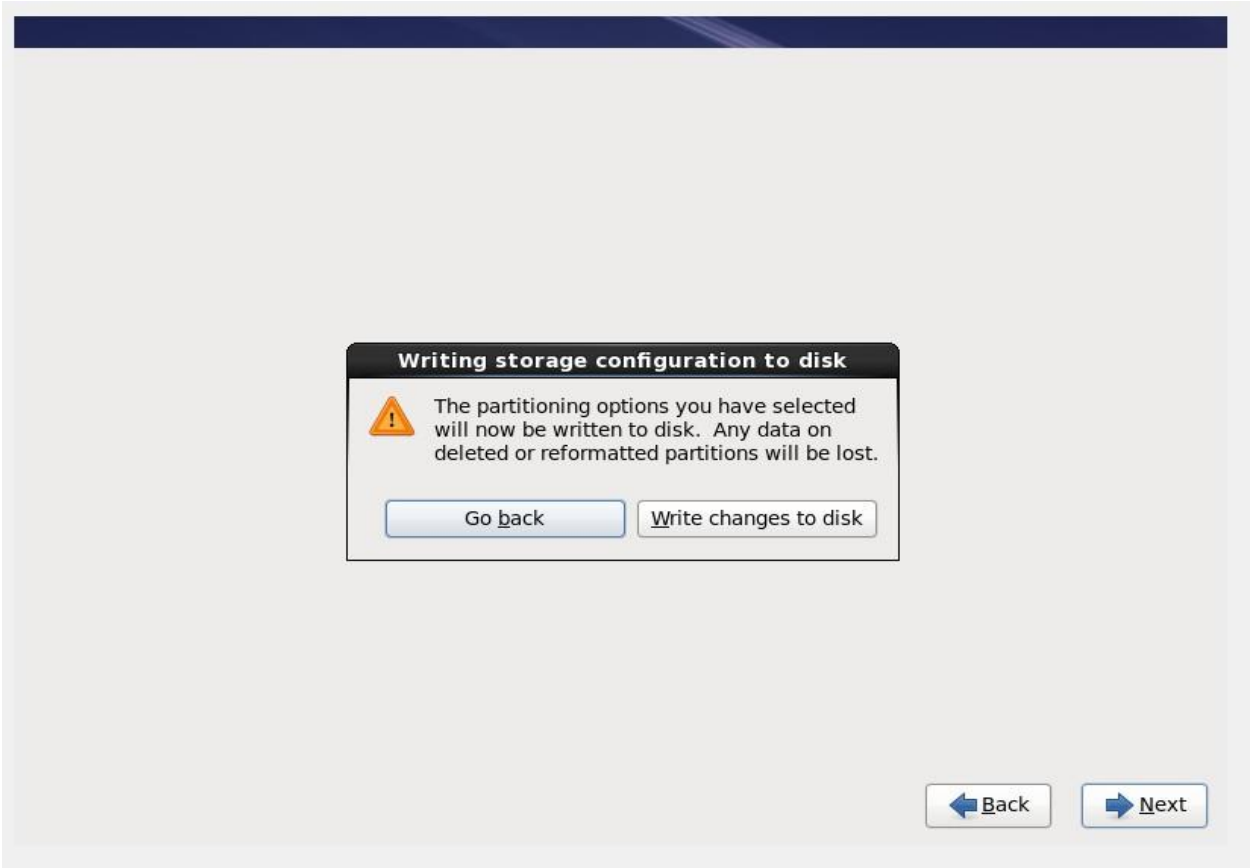
12. You'll also need to specify whether your computer is using UTC for its internal clock. UTC is Coordinated Universal Time, a time standard by which the world regulates clocks and time. It is one of several successors to Greenwich Mean Time, without Daylight Saving Time settings. Most servers have their hardware clocks set to UTC, but most PCs don't. If the hardware clock is set to UTC, the server uses the time zone settings to calculate the local software time. If your computer has its hardware clock set to UTC, select the option System Clock Uses UTC, and click Next to continue. If
not, deselect this option and proceed with the installation.

13. Next you'll specify the password that is to be used by the user root. The root account is used for system administration tasks, and its possibilities are nearly unlimited. Therefore, you should set the root password to something that's not easy for pos- sible intruders to guess.



14. The next screen you'll see is used to specify how you'd like to use the storage
devices on which you'll install Red Hat Enterprise Linux. If you want to go for the easiest solution, select Use All Space. This will remove everything currently installed on the selected hard disk (which typically isn't a bad idea anyway). Table 1.1 gives an overview of all the available options.


Which type of installation would you like?
Use All Space
Removes all partitions on the selected device(s). This includes partitions created by other operating systems.
Tip: This option will remove data from the selected device(s). Make sure you have backups.
Replace Existing Linux System(s)
Removes only Linux partitions (created from a previous Linux installation). This does not remove other partitions you may have on your storage device(s) (such as VFAT or FAT32).
Tip: This option will remove data from the selected device(s). Make sure you have backups.
Shrink Current System
Shrinks existing partitions to create free space for the default layout.
Use Free Space
Retains your current data and partitions and uses only the unpartitioned space on the selected device (s), assuming you have enough free space available.
Create Custom Layout
Manually create your own custom layout on the selected device(s) using our partitioning tool.
Encrypt system
Review and modify partitioning layout
Back
Next


Writing storage configuration to disk
The partitioning options you have selected will now be written to disk. Any data on deleted or reformatted partitions will be lost.
Go back
Write changes to disk
Back
Next

PRACTICAL 1

Working with users, groups and permissions

1. Creating and Managing Users
2. Creating and Managing Groups
3. Using Permissions and Advanced Permissions

1. **Creating and Managing Users**

useradd : Adds a new user into the local authentication database

```
root@localhost:~# su -
Last login: Thu Sep 11 12:40:58 IST 2025 on pts/0
root@localhost:~# cd /home
root@localhost:/home# useradd  soul
useradd: user 'soul' already exists
root@localhost:/home# useradd bob
root@localhost:/home# useradd john
root@localhost:/home# useradd nick
root@localhost:/home# ls
atharv  bob  john  nick  soul
root@localhost:/home# ls -ls
total 4
4 drwx------. 14 atharv atharv 4096 Sep 11 12:28 atharv
0 drwx------.  3 bob    bob      78 Sep 11 12:45 bob
0 drwx------.  3 john   john     78 Sep 11 12:45 john
0 drwx------.  3 nick   nick     78 Sep 11 12:45 nick
0 drwx------.  3 soul   soul     78 Sep 11 12:41 soul
root@localhost:/home# passwd bob
```

passwd : Used to assign, modify, or reset a user's password.

```
root@localhost:/home# passwd bob
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
bob@localhost:~$ su - bob
Password:
Last login: Thu Sep 11 12:47:22 IST 2025 on pts/1
bob@localhost:~$
```

usermod -aG :Updates user account settings.

- -a → append (adds the user to extra groups without removing existing ones).
- -G → specify supplementary groups to add the user into.

UID (User ID) :Every Linux user is identified by a unique numeric ID stored in /etc/passwd.

```
bob@localhost:~$ cd /home
bob@localhost:/home$ su -
Password:
Last login: Thu Sep 11 12:44:23 IST 2025 on pts/1
root@localhost:~# cd /home
root@localhost:/home# ls
atharv  bob  john  nick  soul
root@localhost:/home# useradd ashwini
root@localhost:/home# useradd shark
root@localhost:/home# usermod -aG ashwini shark
root@localhost:/home# id shark
uid=1006(shark) gid=1006(shark) groups=1006(shark),1005(ashwini)
root@localhost:/home#
```

whoami : Prints the username of the account currently logged in.

```
root@localhost:/home# whoami
root
root@localhost:/home# who
root     seat0        2025-09-11 12:40 (login screen)
root     tty2         2025-09-11 12:40 (tty2)
root@localhost:/home# w
 12:54:00 up 14 min,  4 users,  load average: 0.07, 0.08, 0.10
USER     TTY        LOGIN@   IDLE   JCPU   PCPU WHAT
root     tty2      12:40   14:02   0.01s  0.01s /usr/libexec/gnome-session-binary
root               12:40   13:29   0.00s  0.31s /usr/lib/systemd/systemd --user
root     pts/0     12:40   10:08   0.01s  0.01s -bash
root     pts/1     12:44    0.00s  0.08s  0.02s w
root@localhost:/home#
```

Last :Reads from /var/log/wtmp and displays details of previous logins, including user, login/logout time, and session length.

```
root@localhost:/home# groups shark
shark : shark ashwini
root@localhost:/home# last
root     tty2         tty2             Thu Sep 11 12:40   still logged in
root     seat0        login screen     Thu Sep 11 12:40   still logged in
reboot   system boot  6.12.0-55.32.1.e Thu Sep 11 12:39   still running
atharv   tty2         tty2             Thu Sep 11 12:29 - crash  (00:10)
atharv   seat0        login screen     Thu Sep 11 12:29 - crash  (00:10)
reboot   system boot  6.12.0-55.32.1.e Thu Sep 11 12:29   still running
atharv   tty2         tty2             Thu Sep 11 12:24 - crash  (00:04)
atharv   seat0        login screen     Thu Sep 11 12:24 - crash  (00:04)
reboot   system boot  6.12.0-55.32.1.e Thu Sep 11 12:23   still running

wtmp begins Thu Sep 11 12:23:34 2025
root@localhost:/home#
```

- **userdel username** : Removes a user entry from /etc/passwd, /etc/shadow, and /etc/group.
- Does **not** delete the user's home directory or files.

```
root@localhost:/home# ls
ashwini  atharv  bob  john  nick  shark  soul
root@localhost:/home# userdel john
root@localhost:/home# userdel -r nick
root@localhost:/home# ls
ashwini  atharv  bob  john  shark  soul
root@localhost:/home#
```

- **passwd -l username** : Locks a user account (disables login).
- **passwd -u username** : Unlocks the account (enables login again).

```
root@localhost:/home# passwd -l atharv
passwd: password changed.
root@localhost:/home# passwd -u atharv
passwd: password changed.
```

- **mkdir -p /data/girish** : Creates a directory (and any required parent directories if missing).
- **ls -ld directory** : Shows ownership, group, and permission details of a directory.

```
root@localhost:/home# mkdir -p /data/atharv
root@localhost:/home# ls -ld/data/atharv
ls: invalid option -- '/'
Try 'ls --help' for more information.
root@localhost:/home# ls -ld /data/atharv
drwxr-xr-x. 2 root root 6 Sep 11 12:56 /data/atharv
root@localhost:/home#
```

2. Creating and Managing Groups

- **groupadd groupname** : Creates a new group.

```
root@localhost:/home# groupadd alexander
root@localhost:/home# getent group
```

```
root@localhost:/home# getent group alexander
alexander:x:1009:
root@localhost:/home# ls -ltr
total 8
drwx------.  3 soul    soul      78 Sep 11 12:41 soul
drwx------.  3    1003    1003   78 Sep 11 12:45 john
drwx------.  3 ashwini ashwini   78 Sep 11 12:52 ashwini
drwx------.  3 shark   shark     78 Sep 11 12:52 shark
drwx------. 14 bob     bob     4096 Sep 11 13:40 bob
drwx------.  3 nick    soul      78 Sep 11 13:44 nick
drwx------. 14 atharv  atharv  4096 Sep 11 13:51 atharv
-rw-rw-r-x.  1 root    root       0 Sep 11 13:52 testfile.txt
root@localhost:/home#
```

- **ls -ltr** : Lists files in detailed format, sorted by time (oldest first).

```
root@localhost:~# usermod -aG  nick bob
root@localhost:~# ls -ltr
total 4
-rw-------. 1 root root 674 Sep 11 12:23 anaconda-ks.cfg
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Videos
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Templates
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Public
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Pictures
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Music
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Downloads
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Documents
drwxr-xr-x. 2 root root   6 Sep 11 12:40 Desktop
root@localhost:~# cd /home
```

```
root@localhost:/home# id nick
uid=1007(nick) gid=1008(nick) groups=1008(nick)
root@localhost:/home# gpasswd -d nick bob
Removing user nick from group bob
gpasswd: user 'nick' is not a member of 'bob'
root@localhost:/home# groups bob
bob : bob nick
root@localhost:/home# groups nick
nick : nick
root@localhost:/home# usermod -g soul nick
root@localhost:/home# id soul
uid=1001(soul) gid=1001(soul) groups=1001(soul),1002(bob)
root@localhost:/home#
```

- **Usermod -G groupname username** : Assigns a user to specified groups, replacing any existing supplementary groups.

```
root@localhost:/home# usermod -G bob,shark,nick
Usage: usermod [options] LOGIN

Options:
  -a, --append                  append the user to the supplemental GROUPS
                                mentioned by the -G option without removing
                                the user from other groups
  -b, --badname                 allow bad names
  -c, --comment COMMENT         new value of the GECOS field
  -d, --home HOME_DIR           new home directory for the user account
  -e, --expiredate EXPIRE_DATE  set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE       set password inactive after expiration
                                to INACTIVE
  -g, --gid GROUP               force use GROUP as new primary group
  -G, --groups GROUPS           new list of supplementary GROUPS
  -h, --help                    display this help message and exit
  -l, --login NEW_LOGIN         new value of the login name
  -L, --lock                    lock the user account
  -m, --move-home               move contents of the home directory to the
                                new location (use only with -d)
  -o, --non-unique              allow using duplicate (non-unique) UID
  -p, --password PASSWORD       use encrypted password for the new password
  -P, --prefix PREFIX_DIR       prefix directory where are located the /etc/* files
  -r, --remove                  remove the user from only the supplemental GROUPS
                                mentioned by the -G option without removing
                                the user from other groups
  -R, --root CHROOT_DIR         directory to chroot into
  -s, --shell SHELL             new login shell for the user account
```

3. File Permissions and Advanced Permissions

- **ls -l** : Displays files in long format, showing permissions, owner, group, size, and modification time.

```
root@localhost:/home# ls -l
total 8
drwx------.  3 ashwini ashwini   78 Sep 11 12:52 ashwini
drwx------. 14 atharv  atharv  4096 Sep 11 12:28 atharv
drwx------. 14 bob     bob     4096 Sep 11 13:40 bob
drwx------.  3    1003    1003   78 Sep 11 12:45 john
drwx------.  3 nick    soul      78 Sep 11 13:44 nick
drwx------.  3 shark   shark     78 Sep 11 12:52 shark
drwx------.  3 soul    soul      78 Sep 11 12:41 soul
```

- **ls** : Simply lists the contents of the current (or specified) directory.

```
root@localhost:/home# ls
ashwini  atharv  bob  john  nick  shark  soul
```

```
root@localhost:/home# touch /home/atharv/atharvfile.txt
root@localhost:/home# ls -ltr /home/atharv/atharvfile.txt
-rw-r--r--. 1 root root 0 Sep 11 13:51 /home/atharv/atharvfile.txt
root@localhost:/home#
```

- **Chmod**: Adjusts file/directory permissions for user, group, and others (read r, write w, execute x).

```
root@localhost:/home# touch testfile.txt
root@localhost:/home# chmod 444 testfile.txt
root@localhost:/home# ls -l testfile.txt
-r--r--r--. 1 root root 0 Sep 11 13:52 testfile.txt
root@localhost:/home# chmod u+w testfile.txt
root@localhost:/home# ls -l testfile.txt
-rw-r--r--. 1 root root 0 Sep 11 13:52 testfile.txt
root@localhost:/home#
root@localhost:/home# ls -l testfile.txt
-rw-r--r--. 1 root root 0 Sep 11 13:52 testfile.txt
root@localhost:/home# chmod g+w,o+x testfile.txt
root@localhost:/home# ls -l testfile.txt
-rw-rw-r-x. 1 root root 0 Sep 11 13:52 testfile.txt
```

```
root@localhost:/home# ls -l /home/testfile.txt
-rw-rw-r-x. 1 root root 0 Sep 11 13:52 /home/testfile.txt
root@localhost:/home# chown ashwini /home/atharv/atharvfile.txt
root@localhost:/home# ls -l /home/atharvfile.txt
ls: cannot access '/home/atharvfile.txt': No such file or directory
root@localhost:/home# ls -l /home/atharv/atharvfile.txt
-rw-r--r--. 1 ashwini root 0 Sep 11 13:51 /home/atharv/atharvfile.txt
root@localhost:/home# chgrp soul /home/ashwini/ashwinifile.txt
chgrp: cannot access '/home/ashwini/ashwinifile.txt': No such file or directory
root@localhost:/home# chgrp soul /home/atharv/atharvfile.txt
root@localhost:/home# ls -l /home/atharv/atharvfile.txt
-rw-r--r--. 1 ashwini soul 0 Sep 11 13:51 /home/atharv/atharvfile.txt
root@localhost:/home#
```

- chgrp groupname filename : Changes the group ownership of a file or directory.

Practical 2

Aim: Configure password aging policies for a Linux user account.

Background Information: Password Aging Policies are critical security mechanisms in Linux operating systems used to enforce periodic password changes, reduce the window of vulnerability from compromised credentials, and maintain system integrity. In Red Hat Enterprise Linux (RHEL), password lifespan and account expiration rules are managed at two distinct levels:

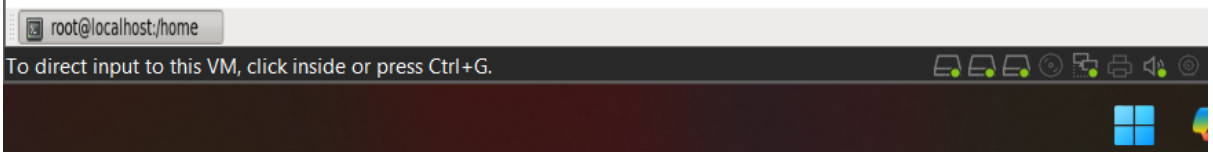
- **User-Specific Management (`chage` command):** A utility used by system administrators to view, set, or modify password aging and expiration properties for individual user accounts.
- **System-Wide Defaults (`/etc/login.defs` file):** A configuration file that stores global default parameters automatically applied to all newly created user accounts on the system.

Key Parameters & Terminology:

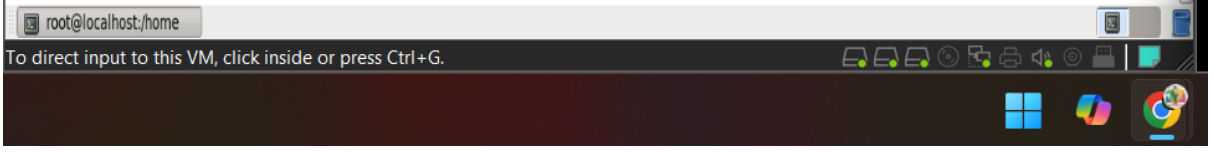
- **Minimum Password Age (`-m`):** Specifies the minimum number of days that must pass before a user is allowed to change their password again, preventing rapid cycling to bypass password history restrictions.
- **Maximum Password Age (`-M`):** Defines the maximum number of days a password remains valid. Once this period expires, the user is forced to change their password upon login.
- **Warning Days (`-W`):** Specifies the number of days prior to password expiration during which a warning message is displayed to the user at login.
- **Inactive Days (`-I`):** Specifies the grace period (in days) after a password expires during which the user can still update it. If the password is not changed within this window, the account is set to an inactive/locked state.
- **Account Expiration Date (`-E`):** Sets an absolute calendar date (in `YYYY-MM-DD` format) on which the account is disabled entirely, regardless of the password's expiration status.



```
[root@localhost home]# chage -m 1 -M 3 -W 1 -I 5 user2
[root@localhost home]# chage -l user2
Last password change                                    : Sep 18, 2026
Password expires                                        : Sep 21, 2026
Password inactive                                       : Sep 26, 2026
Account expires                                         : never
Minimum number of days between password change          : 1
Maximum number of days between password change          : 3
Number of days of warning before password expires       : 1
[root@localhost home]# S
```




```
[root@localhost home]# chage -E 2026-09-20 user2
[root@localhost home]# chage -l user2
Last password change                                    : Sep 18, 2026
Password expires                                        : Sep 21, 2026
Password inactive                                       : Sep 26, 2026
Account expires                                         : Sep 20, 2026
Minimum number of days between password change          : 1
Maximum number of days between password change          : 3
Number of days of warning before password expires       : 1
[root@localhost home]#
```



Practical 3

Aim: Modify the ownership of a file and validate the changes

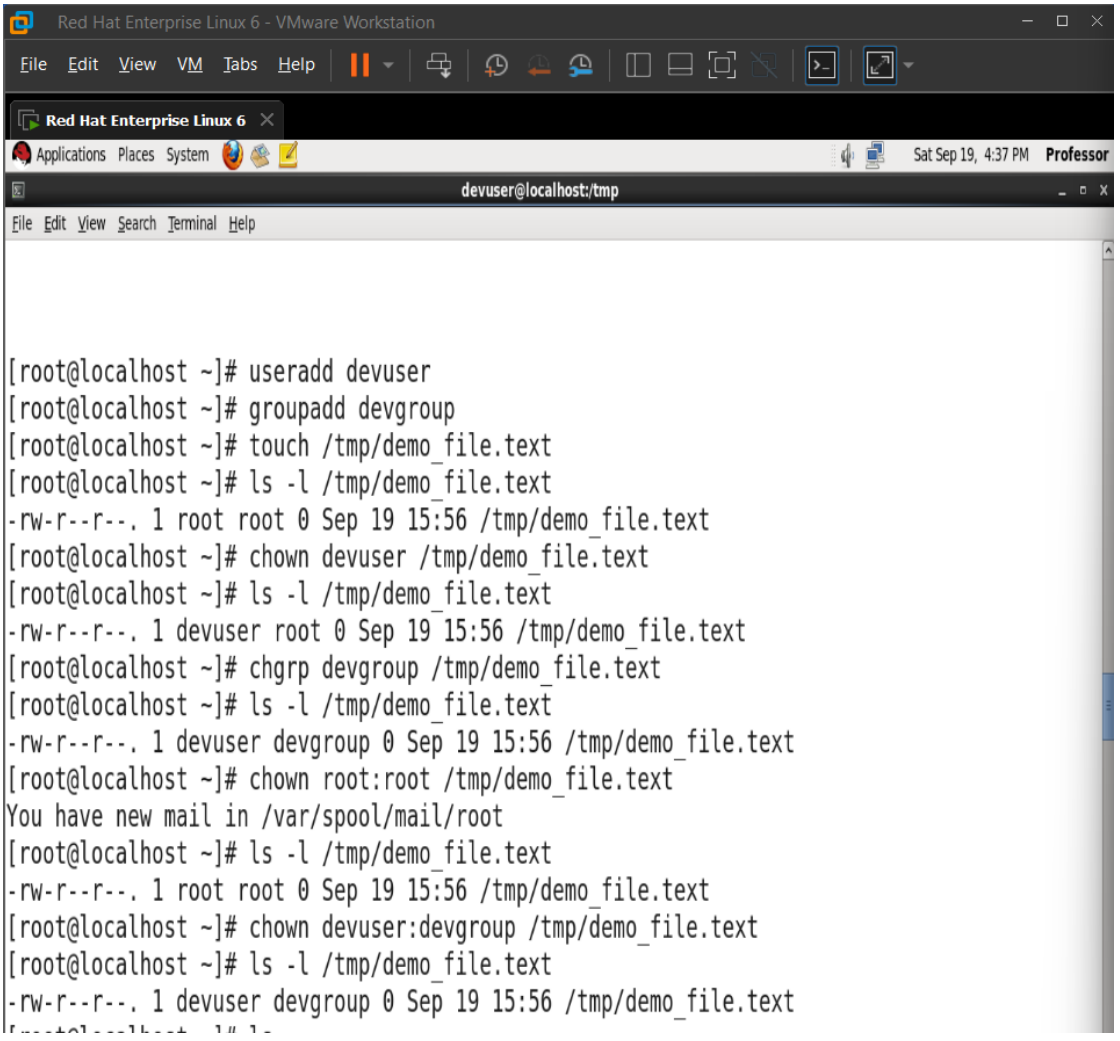
Background Information

In a Linux multi-user environment, to maintain security, every file and directory belongs to a User Owner and a Group Owner.

- **User Owner:** The individual user who has primary access and control over the file.
- **Group Owner:** A group of users whose members have shared access permissions to the file.

Key Commands:

- **chown (Change Owner):** This command is used to change the User Owner, Group Owner, or both for a file simultaneously.
- **chgrp (Change Group):** This command is used to change specific group ownership.
- **ls -l (Long Listing):** Used to view file metadata, owner name, group name, and permissions to validate changes.



Practical 4

Aim: Configure and verify Access Control Lists (ACLs) for files and directories

Background Information

Standard Linux file permissions (UGO - User, Group, Other) have a limitation: you can set permissions for only one Owner User and one Owner Group on a file. If you need to grant distinct permissions to a third specific user, the standard `chmod` command is insufficient.

ACL (Access Control List) provides flexible permission control, allowing you to assign exact permissions (`rwx`) to any individual user or group.

Key Commands:

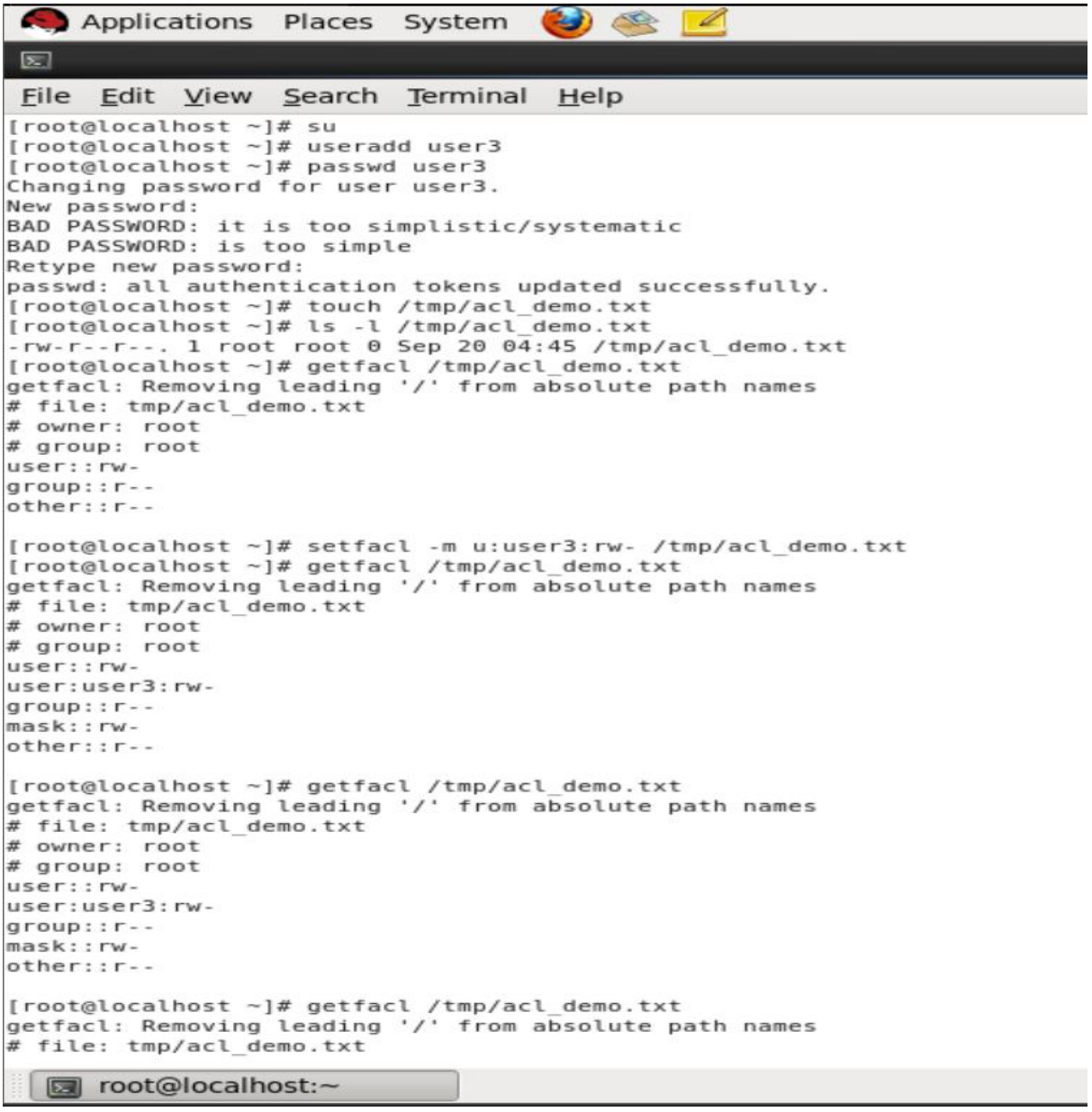
- **getfacl:** Used to view the ACL settings applied to a file or directory.
- **setfacl:** Used to modify, assign, or remove ACL permissions on a file or directory.

Important Flags:

- **-m:** Used to modify or create ACL entries (e.g., `u:username:rwx` or `g:groupname:rwx`).
- **-x:** Used to remove an ACL entry for a specific user or group.
- **-b:** Used to remove all ACL entries at once from a file or directory and reset them to standard permissions.
- **-R:** Used to recursively apply ACL permissions to all files and subdirectories within a directory.

Identification:

When ACL is enabled on a file or directory, a plus sign (`+`) appears at the end of the permission string in the `ls -l` command output (e.g., `-rw-r-----+`).


Applications Places System
File Edit View Search Terminal Help
[root@localhost ~]# su
[root@localhost ~]# useradd user3
[root@localhost ~]# passwd user3
Changing password for user user3.
New password:
BAD PASSWORD: it is too simplistic/systematic
BAD PASSWORD: is too simple
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# touch /tmp/acl_demo.txt
[root@localhost ~]# ls -l /tmp/acl_demo.txt
-rw-r--r--. 1 root root 0 Sep 20 04:45 /tmp/acl_demo.txt
[root@localhost ~]# getfacl /tmp/acl_demo.txt
getfacl: Removing leading '/' from absolute path names
file: tmp/acl_demo.txt
owner: root
group: root
user::rw-
group::r--
other::r--
[root@localhost ~]# setfacl -m u:user3:rw- /tmp/acl_demo.txt
[root@localhost ~]# getfacl /tmp/acl_demo.txt
getfacl: Removing leading '/' from absolute path names
file: tmp/acl_demo.txt
owner: root
group: root
user::rw-
user:user3:rw-
group::r--
mask::rw-
other::r--
[root@localhost ~]# getfacl /tmp/acl_demo.txt
getfacl: Removing leading '/' from absolute path names
file: tmp/acl_demo.txt
owner: root
group: root
user::rw-
user:user3:rw-
group::r--
mask::rw-
other::r--
[root@localhost ~]# getfacl /tmp/acl_demo.txt
getfacl: Removing leading '/' from absolute path names
file: tmp/acl_demo.txt
root@localhost:~


[root@localhost ~]# ls -l /tmp/acl_demo.txt
-rw-rw-r--+ 1 root root 0 Sep 20 04:45 /tmp/acl_demo.txt
[root@localhost ~]# groupadd devteam
groupadd: group 'devteam' already exists
[root@localhost ~]# setfacl -m g:devteam:rwx /tmp/acl_demo.txt
[root@localhost ~]# getfacl /tmp/acl_demo.txt
getfacl: Removing leading '/' from absolute path names
file: tmp/acl_demo.txt
owner: root
group: root
user::rw-
user:user3:rw-
group::r--
group:devteam:rwx
mask::rwx
other::r--
[root@localhost ~]# setfacl -x u:user3 /tmp/acl_demo.txt
[root@localhost ~]# getfacl /tmp/acl_demo.txt
getfacl: Removing leading '/' from absolute path names
file: tmp/acl_demo.txt
owner: root
group: root
user::rw-
group::r--
group:devteam:rwx
mask::rwx
other::r--
[root@l root@localhost:~

Practical 5

Aim: Background Information

Whenever you create a new file or directory in Linux, the system automatically assigns certain default permissions to it. `umask` (User Mask) is used to determine these default permissions.

Default Base Permissions (Before Umask is Applied):

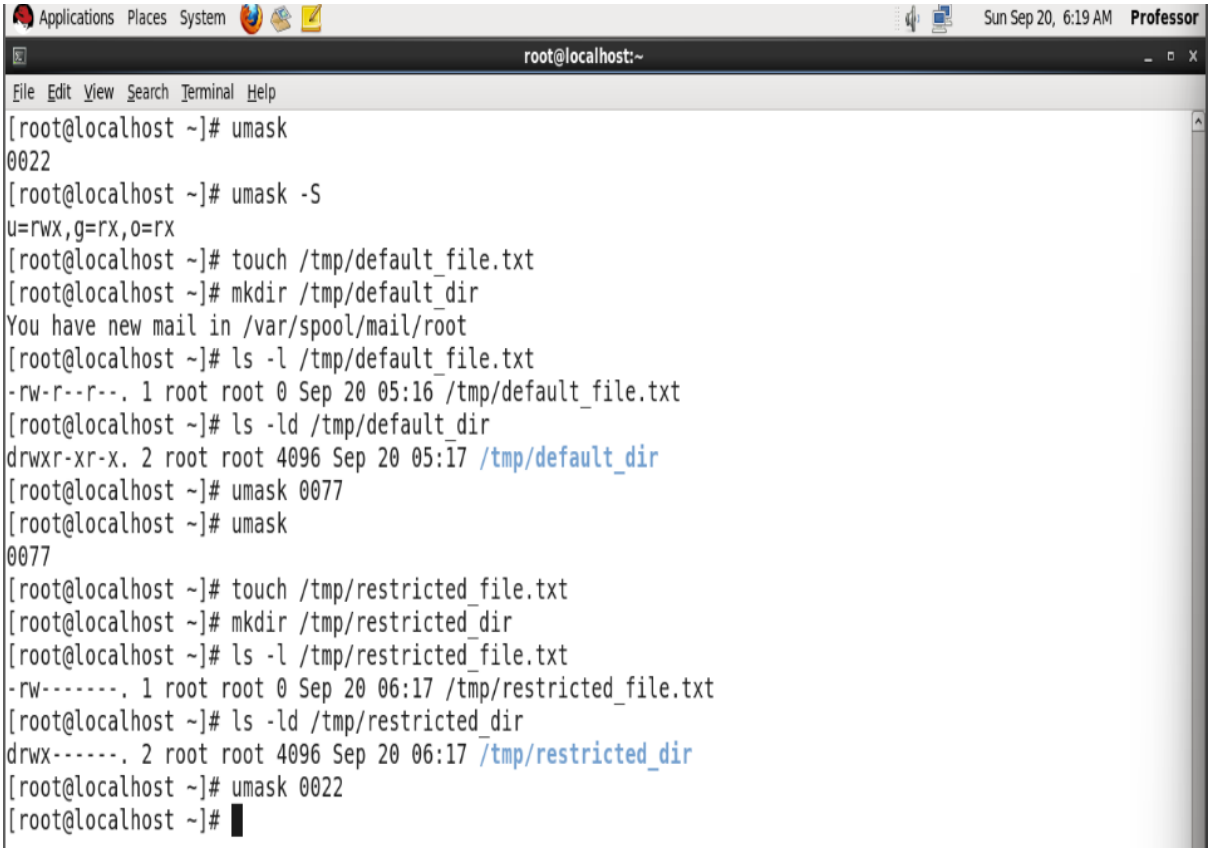
- **For Files:** `666` (`rw-rw-rw-`) — Execution permission (`x`) is not granted to files by default for security reasons.
- **For Directories:** `777` (`rwxrwxrwx`) — Execution permission is mandatory to open or navigate a directory.

Permission Calculation Formula:

- **File Permission** = `666 - umask value`
- **Directory Permission** = `777 - umask value`

Default Umask Values in RHEL6:

- **Root User:** `0022` $\rightarrow$ File permissions: `644` (`rw-r--r--`), Directory permissions: `755` (`rwxr-xr-x`).
- **Regular User:** `0002` $\rightarrow$ File permissions: `664` (`rw-rw-r--`), Directory permissions: `775` (`rwxrwxr-x`).



Practical 6

Aim: Inspect, modify, and validate file attributes in the Linux operating system

Background Information In addition to standard Linux file permissions (`chmod`) and Access Control Lists (ACL), ext2, ext3, and ext4 file systems provide the option of **Special File Attributes**. These attributes offer high-level security at the file system level.

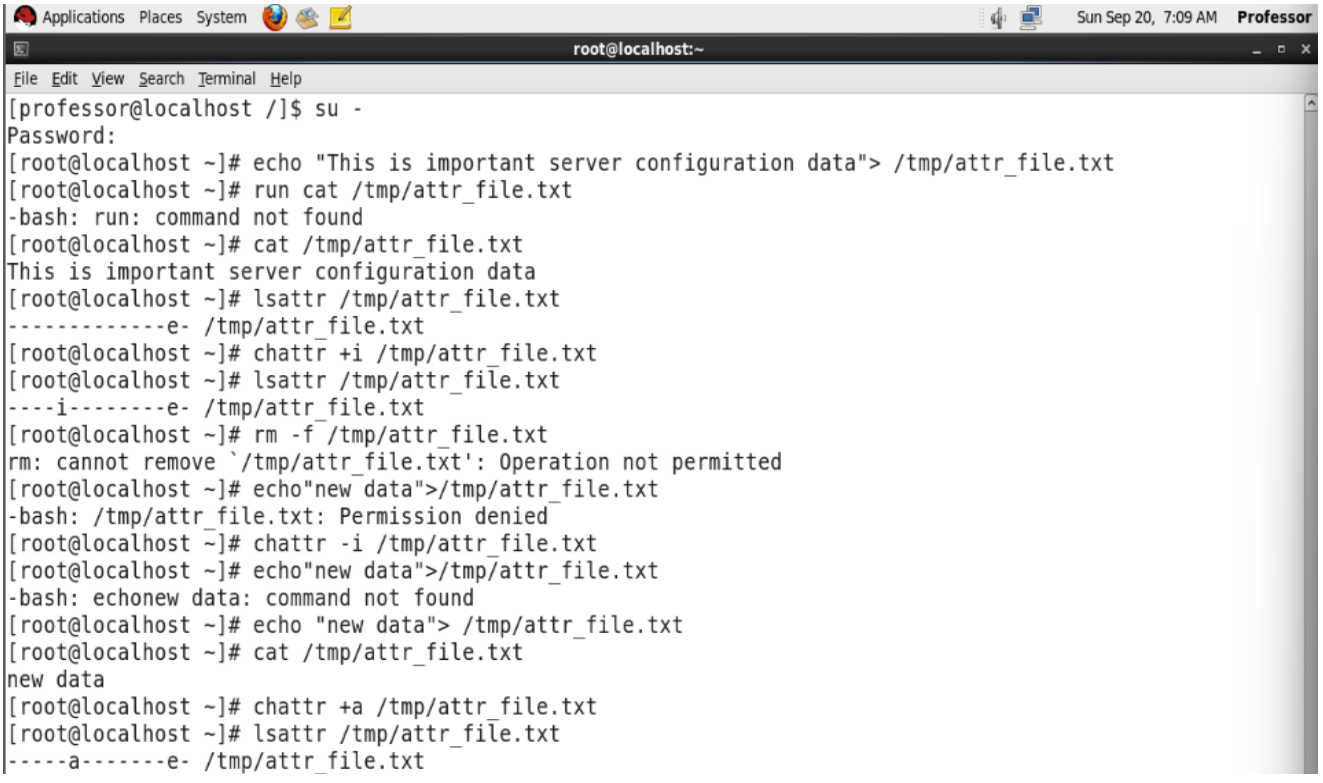
Speciality: If the Immutable attribute (`+i`) is applied to a file, even the **root user cannot modify, rename, truncate, or delete** that file until the attribute is explicitly removed!

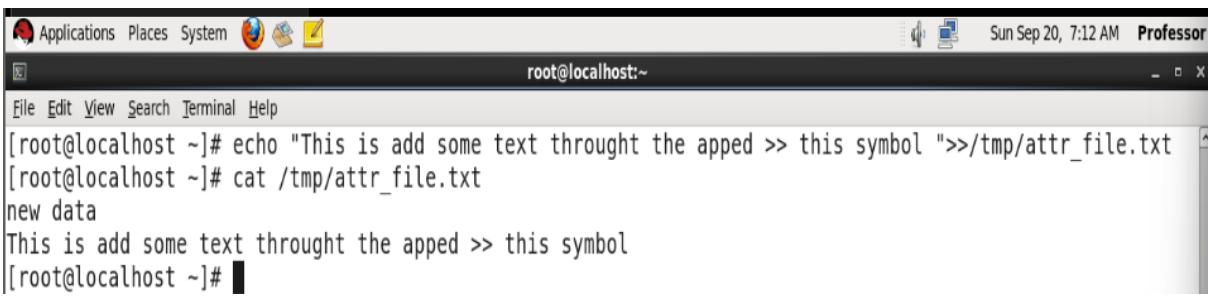
Main Attributes:

- **`i` (Immutable):** Completely freezes or locks the file. Editing, appending, deleting, or renaming is not allowed.
- **`a` (Append-Only):** Allows new data to be appended only at the end (`>>`) of the file. Existing content cannot be deleted or overwritten (`>`) (ideal for log files).

Key Commands:

- **`lsattr` (List Attributes):** Used to check and inspect active attributes on a file or directory.
- **`chattr` (Change Attributes):** Used to set (`+`), remove (`-`), or assign (`=`) attributes on a file or directory.





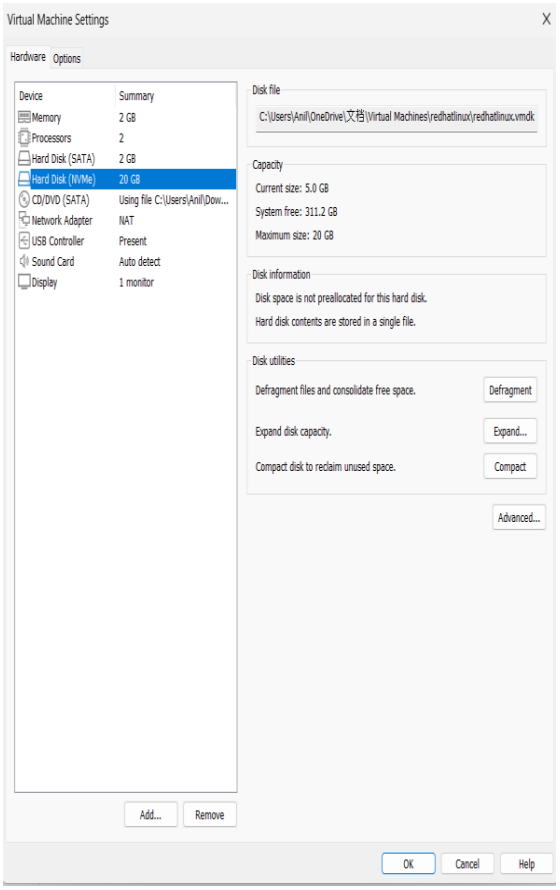
PRACTICAL 7

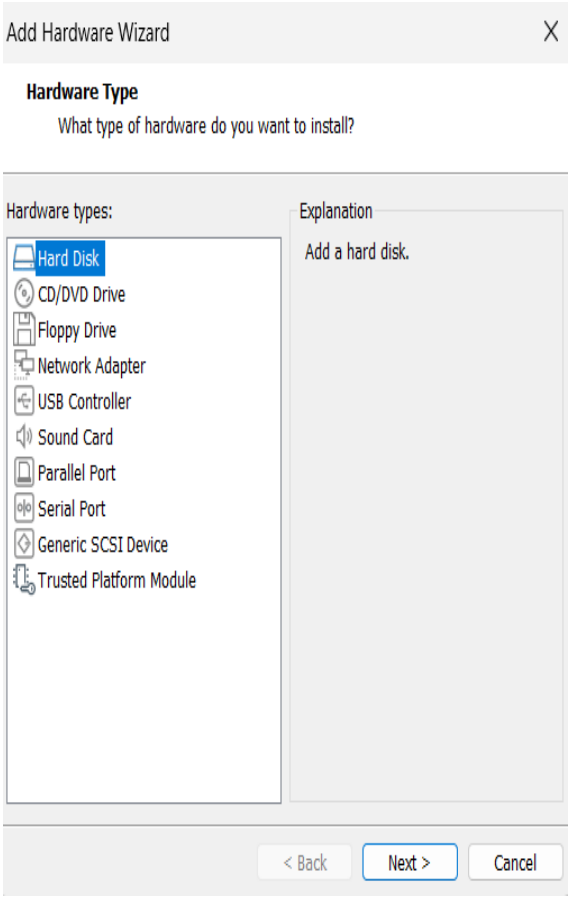
Perform the following steps to create an LVM logical volume on a Linux system:

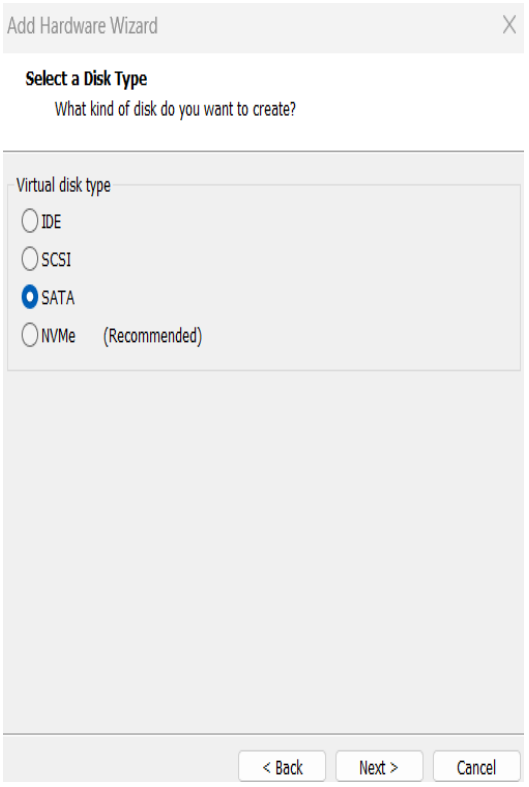
a. Install a new Hard Disk drive
b. Make a partition to use it
c. Designate physical volume (PV)
d. Manage Volume Group (VG)
e. Manage Logical Volume (LV)
f. Apply a filesystem
g. Set a mount point

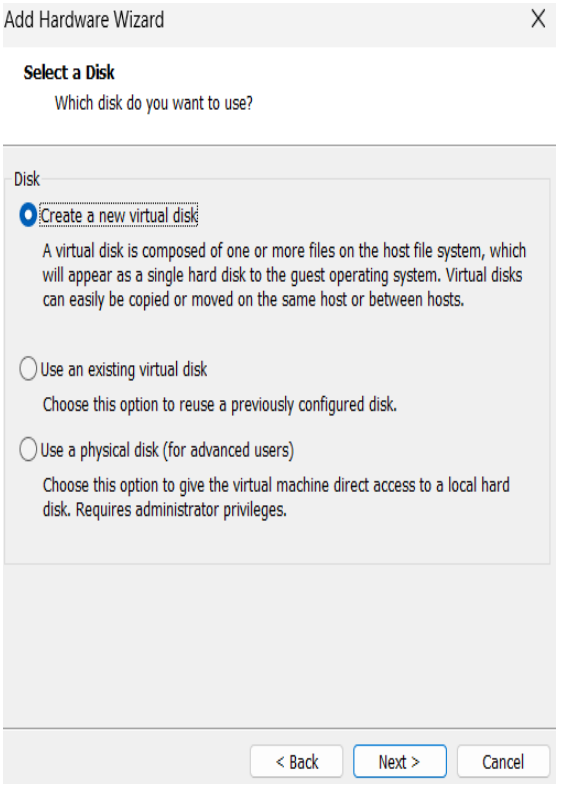
--LVM- In Linux, LVM stands for Logical Volume Manager.It is a system used for managing disk storage more flexibly than traditional partitioning.

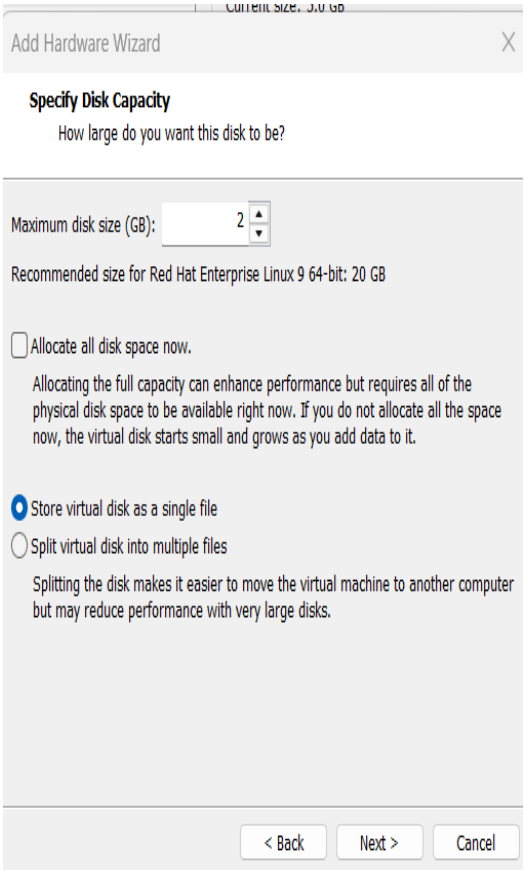
Steps to create new hard disk:

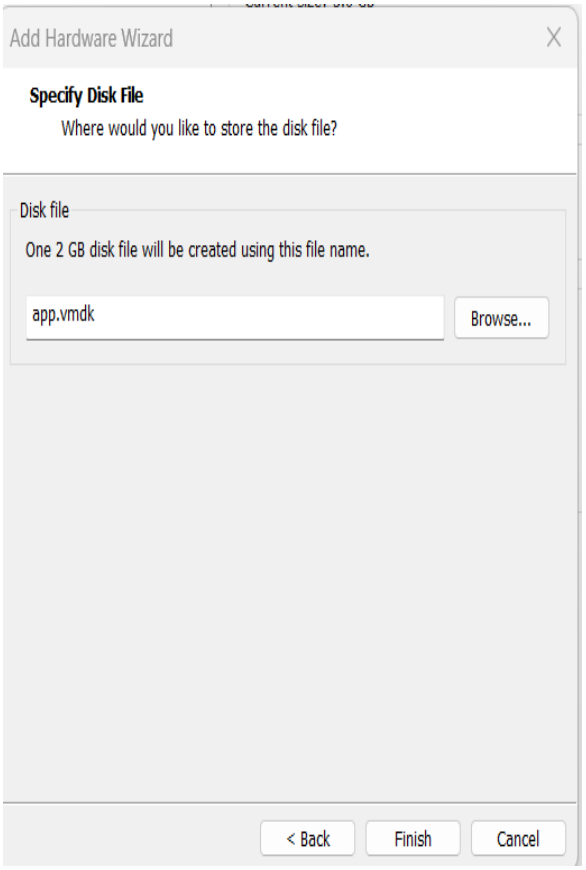





Add Hardware Wizard
Select a Disk Type
What kind of disk do you want to create?
Virtual disk type
IDE
SCSI
SATA
NVMe (Recommended)
< Back
Next >
Cancel


Add Hardware Wizard
Select a Disk
Which disk do you want to use?
Disk
Create a new virtual disk
A virtual disk is composed of one or more files on the host file system, which will appear as a single hard disk to the guest operating system. Virtual disks can easily be copied or moved on the same host or between hosts.
Use an existing virtual disk
Choose this option to reuse a previously configured disk.
Use a physical disk (for advanced users)
Choose this option to give the virtual machine direct access to a local hard disk. Requires administrator privileges.
< Back
Next >
Cancel


Add Hardware Wizard
Specify Disk Capacity
How large do you want this disk to be?
Maximum disk size (GB): 2
Recommended size for Red Hat Enterprise Linux 9 64-bit: 20 GB
Allocate all disk space now.
Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
< Back
Next >
Cancel


Add Hardware Wizard
Specify Disk File
Where would you like to store the disk file?
Disk file
One 2 GB disk file will be created using this file name.
app.vmdk
Browse...
< Back
Finish
Cancel

a. Install a new Hard Disk drive

fdisk -l: Lists all available disks and their partitions on a Linux system.

```
redhatlinux@localhost:~$ su -
Password:
Last login: Sat Aug 30 21:35:15 IST 2025 on pts/0
root@localhost:~# fdisk -l
Disk /dev/nvme0n1: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 65A0F57F-EFC0-4F45-A3E2-28702E92631D

Device           Start      End  Sectors  Size Type
/dev/nvme0n1p1    2048  1230847  1228800  600M EFI System
/dev/nvme0n1p2 1230848  3327999  2097152    1G Linux extended boot
/dev/nvme0n1p3 3328000 41940991 38612992 18.4G Linux LVM


Disk /dev/sda: 2 GiB, 2147483648 bytes, 4194304 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/rhel-root: 16.41 GiB, 17620271104 bytes, 34414592 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

b. Make a partition to use it

fdisk /dev/sda: fdisk /dev/sda → Opens the partitioning tool fdisk to create, delete, or manage partitions on the /dev/sda disk

```
root@localhost:~# fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xd497e049.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194303, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-4194303, default 4194303):

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

c. Designate physical volume (PV)

pvcreate: This command is used to create physical volume.
pvdisplay: Shows detailed information about all Physical Volumes (PVs) in LVM, such as size, UUID, and volume group association.

```
root@localhost:~# pvdisplay
  --- Physical volume ---
  PV Name               /dev/nvme0n1p3
  VG Name               rhel
  PV Size               18.41 GiB / not usable 2.00 MiB
  Allocatable           yes (but full)
  PE Size               4.00 MiB
  Total PE              4713
  Free PE               0
  Allocated PE          4713
  PV UUID               ycN6mg-VUVm-91se-YQsD-kow0-S4IC-ZW9xfp
```

d. Manage Volume Group (VG)

```
root@localhost:~# vgcreate vgdata /dev/sda1
  Volume group "vgdata" successfully created
root@localhost:~# vgdisplay
  --- Volume group ---
  VG Name               rhel
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  3
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                2
  Open LV               2
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               18.41 GiB
  PE Size               4.00 MiB
  Total PE              4713
  Alloc PE / Size       4713 / 18.41 GiB
  Free  PE / Size       0 / 0
  VG UUID               VtBbUc-o4a1-PVAT-UfDf-52kl-e0xe-f6YSH2
```

e. Manage Logical Volume (LV)

lvcreate -L 500M -n lvdata vgdata

```
root@localhost:~# mkfs.ext4 /dev/vgdata/lvdata
mke2fs 1.47.1 (20-May-2024)
The file /dev/vgdata/lvdata does not exist and no size was specified.
root@localhost:~# lvcreate -L 500M -n lvdata vgdata
  Logical volume "lvdata" created.
root@localhost:~# mkfs.ext4 /dev/vgdata/lvdata
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 512000 1k blocks and 128016 inodes
Filesystem UUID: 6f2c5ae4-2167-4c6f-bec5-6bf8e240c8fd
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

```
root@localhost:~# lvdisplay
  --- Logical volume ---
  LV Path                /dev/rhel/swap
  LV Name                swap
  VG Name                rhel
  LV UUID                MNHR65-xsnQ-E2BA-NZr5-TPG1-GSLY-gkjt2J
  LV Write Access        read/write
  LV Creation host, time localhost.localdomain, 2025-08-30 10:46:39 +0530
  LV Status              available
  # open                 1
  LV Size                2.00 GiB
  Current LE             512
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:1

  --- Logical volume ---
  LV Path                /dev/rhel/root
  LV Name                root
  VG Name                rhel
  LV UUID                yW0331-Hu7G-HGV5-Sdxe-KkQQ-po3A-pgNX7U
  LV Write Access        read/write
  LV Creation host, time localhost.localdomain, 2025-08-30 10:46:39 +0530
  LV Status              available
  # open                 1
  LV Size                16.41 GiB
  Current LE             4201
  Segments               1
  Allocation             inherit
```

f. Set a mount point

Create a directory and mount the lv
mkdir /mnt/lvdata
mount /dev/vgdata/lvdata /mnt/lvdata df -h

```
root@localhost:~# mkdir /mnt/lvdata
mkdir: cannot create directory '/mnt/lvdata': File exists
root@localhost:~# mkdir /mnt/lvdata1
root@localhost:~# del mkdir /mnt/lvdata
bash: del: command not found...
root@localhost:~# sudo rm -r /mnt/lvdata
root@localhost:~# mkdir /mnt/lvdata
root@localhost:~# mount /dev/vgdata/lvdata /mnt/lvdata
root@localhost:~# df -h
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root      17G  4.1G   13G  25% /
devtmpfs                  4.0M     0  4.0M   0% /dev
tmpfs                     836M   84K  836M   1% /dev/shm
efivarfs                  256K   56K  196K  23% /sys/firmware/efi/efivars
tmpfs                     335M  6.9M  328M   3% /run
tmpfs                     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2            960M  326M  635M  34% /boot
/dev/nvme0n1p1            599M  8.4M  591M   2% /boot/efi
tmpfs                     168M  140K  167M   1% /run/user/1000
/dev/sr0                  7.9G  7.9G     0 100% /run/media/redhatlinux/RHEL-10-0-BaseOS-x86_64
tmpfs                     168M   60K  168M   1% /run/user/0
/dev/mapper/vgdata-lvdata  459M   14K  430M   1% /mnt/lvdata
```

f. Apply a filesystem

Make the Mount Permanent
Edit the /etc/fstab file so that the LV mounts automatically after reboot: vi /etc/fstab
In file, add this line at end:
/dev/vgdata/lvdata /mnt/lvdata ext4 defaults 0 0

```
root@localhost:~# vi /etc/fstab
root@localhost:~# mount -a
root@localhost:~# df -h
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root      17G  4.1G   13G  25% /
devtmpfs                  4.0M     0  4.0M   0% /dev
tmpfs                     836M   84K  836M   1% /dev/shm
efivarfs                  256K   56K  196K  23% /sys/firmware/efi/efivars
tmpfs                     335M  6.9M  328M   3% /run
tmpfs                     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2            960M  326M  635M  34% /boot
/dev/nvme0n1p1            599M  8.4M  591M   2% /boot/efi
tmpfs                     168M  140K  167M   1% /run/user/1000
/dev/sr0                  7.9G  7.9G     0 100% /run/media/redhatlinux/RHEL-10-0-BaseOS-x86_64
tmpfs                     168M   60K  168M   1% /run/user/0
/dev/mapper/vgdata-lvdata  459M   14K  430M   1% /mnt/lvdata
```

Practical no: 8

Objective: **Create Snapshots in LVM on Linux.**

Step 1: Check the existing LVM's

Use lvdisplay

```
[root@localhost ~]# lvdisplay
  --- Logical volume ---
  LV Name                /dev/vgapps/app1-lv
  VG Name                vgapps
  LV UUID                pcd8ku-i9IY-cPNw-Xx7g-2wyF-Ll2S-XTv2Rx
  LV Write Access        read/write
  LV Status              available
  # open                 1
  LV Size                100.00 MiB
  Current LE             25
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:2

  --- Logical volume ---
  LV Name                /dev/vgapps/app-lv
  VG Name                vgapps
  LV UUID                dKFpKH-j4BH-nV0R-aRQF-dDJp-W2Nw-3i1lGf
  LV Write Access        read/write
  LV Status              available
  # open                 1
```

Step 2: Create a Snapshot in LV

Lvcreate –size 2M –snapshot –name vol1_snapshot /dev/vgapps/app-lv

```
[root@localhost ~]# lvcreate --size 2M --snapshot --name vol1_snapshot /dev/vgap
ps/app-lv1
  Rounding up size to full physical extent 4.00 MiB
  Logical volume "vol1_snapshot" created
[root@localhost ~]#
```

Step3: verify using lvdisplay

```
  --- Logical volume ---
  LV Name                /dev/vgapps/vol1_snapshot
  VG Name                vgapps
  LV UUID                x8IdYC-Omvb-ywII-Z2VG-IkzX-VBkC-DAwJGS
  LV Write Access        read/write
  LV snapshot status     active destination for /dev/vgapps/app-lv1
  LV Status              available
  # open                 0
  LV Size                4.00 MiB
  Current LE             1
  COW-table size         4.00 MiB
  COW-table LE           1
  Allocated to snapshot  0.29%
  Snapshot chunk size    4.00 KiB
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:4
```

Step 3: create a mount point using mkdir /mnt/vol1,

then mount the snapshot using mount/dev/vgapps/vol1_snapshot /mnt/vol1

and check it ls /mnt/vol1

```
 --- Logical volume ---
 LV Name                /dev/vgapps/vol1_snapshot
 VG Name                vgapps
 LV UUID                x8IdYC-Omvb-ywII-Z2VG-IkzX-VBkC-DAwJGS
 LV Write Access        read/write
 LV snapshot status     active destination for /dev/vgapps/app-lv1
 LV Status              available
 # open                 0
 LV Size                4.00 MiB
 Current LE             1
 COW-table size         4.00 MiB
 COW-table LE           1
 Allocated to snapshot  0.29%
 Snapshot chunk size    4.00 KiB
 Segments               1
 Allocation             inherit
 Read ahead sectors     auto
 - currently set to     256
 Block device           253:4
```

Step 4: unmount it and delete

```
[root@localhost ~]# umount /mnt/vol1
[root@localhost ~]# lvremove /dev/vgapps/vol1_snapshot
Do you really want to remove active logical volume vol1_snapshot? [y/n]: y
  Logical volume "vol1_snapshot" successfully removed
[root@localhost ~]#
```

Step 5: verify using lvdisplay

```
[root@localhost ~]# sysctl vm.swappiness=15
vm.swappiness = 15
[root@localhost ~]# cat /proc/sys/vm/swappiness
15
[root@localhost ~]#
```

```
 LV Status              available
 # open                 1
 LV Size                3.97 GiB
 Current LE             1016
 Segments               1
 Allocation             inherit
 Read ahead sectors     auto
 - currently set to     256
 Block device           253:1

root@localhost ~]#
```

S

Practical no: 9

Steps to Create and Enable Swap File in Linux

Step: 1 -

To check swappiness utilization value

```
[root@localhost ~]# cat /proc/sys/vm/swappiness
60
[root@localhost ~]#
```

To change the swappiness value but it will change temporary

To see the space availability

```
[root@localhost ~]# ls -l /tmp
total 40
drwx------. 2 redhat redhat 4096 Jun 13 11:57 keyring-9IiIrI
drwx------. 2 redhat redhat 4096 Jun  3 15:15 keyring-FkTmAg
drwx------. 2 redhat redhat 4096 Jul  2 12:29 keyring-KdXuD8
drwx------. 2 gdm    gdm    4096 Jul  2 12:29 orbit-gdm
drwx------. 2 redhat redhat 4096 Jul  2 12:29 orbit-redhat
drwx------. 2 gdm    gdm    4096 Jul  2 12:29 pulse-KqAtmXwLb8Fr
drwx------. 2 redhat redhat 4096 Jul  2 12:29 pulse-kWPeeGXGyPXY
drwx------. 2 redhat redhat 4096 Jul  2 12:29 virtual-redhat.igrPpa
drwx------. 2 redhat redhat 4096 Jun 13 11:57 virtual-redhat.JYQH2C
drwx------. 2 redhat redhat 4096 Jun  3 15:15 virtual-redhat.znRVqC
-rw-------. 1 root   root      0 Jun  3 14:05 yum.log
[root@localhost ~]#
```

As you see in the image below, the root has much space to create swap. Let's use it

```
[root@localhost ~]# df -hT
Filesystem    Type    Size  Used Avail Use% Mounted on
/dev/mapper/VolGroup-lv_root
              ext4    5.5G  2.3G  3.0G  44% /
tmpfs        tmpfs    504M  476K  504M   1% /dev/shm
/dev/sda1     ext4    485M   29M  431M   7% /boot
/dev/sr0   iso9660    2.8G  2.8G     0 100% /media/RHEL_6.0 i386 Disc 1
[root@localhost ~]#
```

Create file

```
[root@localhost ~]# fallocate -l 1G /swapfile
[root@localhost ~]#
```

Verify this file

```
1048576+0 records out
1073741824 bytes (1.1 GB) copied, 4.1215 s, 261 MB/s
[root@localhost ~]# ls -lh /swapfile2
-rw-r--r--. 1 root root 1.0G Jul  2 20:03 /swapfile2
[root@localhost ~]#
```

2nd method to create a file if fallocate command not work

```
[root@localhost ~]# dd if=/dev/zero of=/swapfile2 bs=1024 count=1048576
1048576+0 records in
1048576+0 records out
1073741824 bytes (1.1 GB) copied, 4.1215 s, 261 MB/s
[root@localhost ~]# ls -lh /swapfile2
-rw-r--r--. 1 root root 1.0G Jul  2 20:03 /swapfile2
[root@localhost ~]#
```

To check the permissions on these files

```
[root@localhost ~]# ls -lh /swapfile*
-rw-r--r--. 1 root root 1.0G Jul  2 19:54 /swapfile
-rw-r--r--. 1 root root 1.0G Jul  2 20:03 /swapfile2
[root@localhost ~]#
```

Change the file permission

```
[root@localhost ~]# chmod 600 /swapfile
[root@localhost ~]# chmod 600 /swapfile2
[root@localhost ~]# ls -lh /swapfile*
-rw-------. 1 root root 1.0G Jul  2 19:54 /swapfile
-rw-------. 1 root root 1.0G Jul  2 20:03 /swapfile2
[root@localhost ~]#
```

Make swap on above swapfile

```
[root@localhost ~]# mkswap /swapfile
mkswap: /swapfile: warning: don't erase bootbits sectors
        on whole disk. Use -f to force.
Setting up swapspace version 1, size = 1048572 KiB
no label, UUID=e5b6b966-1ac5-4c06-8237-43565093fb65
[root@localhost ~]#
```

How to verify

```
[root@localhost ~]# blkid /swapfile
/swapfile: UUID="e5b6b966-1ac5-4c06-8237-43565093fb65" TYPE="swap"
[root@localhost ~]#
```

Make swap for second file

```
[root@localhost ~]# mkswap /swapfile2
Setting up swapspace version 1, size = 1048572 KiB
no label, UUID=74787583-dec2-4d74-8dac-10545ecd18a0
[root@localhost ~]# blkid /swapfile
/swapfile: UUID="e5b6b966-1ac5-4c06-8237-43565093fb65" TYPE="swap"
[root@localhost ~]#
```

Lets own these swap file 1 (swapfile)

```
[root@localhost ~]# swapon /swapfile -v
swapon on /swapfile
swapon: /swapfile: found swap signature: version 1, page-size 4, same byte order
swapon: /swapfile: pagesize=4096, swapsize=1073741824, devsize=1073741824
```

Verify with these twice method

```
[root@localhost ~]# free -m
             total       used       free     shared    buffers     cached
Mem:          2023       1453        570          0         31       1234
-/+ buffers/cache:        187       1835
Swap:         5087          0       5087
[root@localhost ~]# cat /proc/swaps
Filename                                Type            Size    Used    Priority
/dev/dm-1                               partition       4161528 0       -1
/swapfile                               file            1048568 0       -2
[root@localhost ~]#
```

root@localhost:~

Lets own these swap file 2 (swapfile2)

```
[root@localhost ~]# swapon /swapfile2 -v
swapon on /swapfile2
swapon: /swapfile2: found swap signature: version 1, page-size 4, same byte order
swapon: /swapfile2: pagesize=4096, swapsize=1073741824, devsize=1073741824
[root@localhost ~]# free -m
             total       used       free     shared    buffers     cached
Mem:          2023       1453        569          0         31       1234
-/+ buffers/cache:        187       1835
Swap:         6111          0       6111
[root@localhost ~]# cat /proc/swaps
Filename                                Type            Size    Used    Priority
/dev/dm-1                               partition       4161528 0       -1
/swapfile                               file            1048568 0       -2
/swapfile2                              file            1048568 0       -3
[root@localhost ~]#
```

Also can see in the top files

```
top - 21:00:28 up  1:13,  2 users,  load average: 0.00, 0.84, 1.02
Tasks: 133 total,   1 running, 131 sleeping,   1 stopped,   0 zombie
Cpu(s):  0.0%us,  0.0%sy,  0.0%ni,100.0%id,  0.0%wa,  0.0%hi,  0.0%si,  0.0%st
Mem:   2071752k total,  1489532k used,   582220k free,    32004k buffers
Swap:  6258664k total,        0k used,  6258664k free,  1264512k cached

  PID USER      PR  NI  VIRT  RES  SHR S %CPU %MEM    TIME+  COMMAND
    1 root      20   0  2824 1408 1200 S  0.0  0.1   0:00.55 init
    2 root      20   0     0    0    0 S  0.0  0.0   0:00.00 kthreadd
    3 root      RT   0     0    0    0 S  0.0  0.0   0:00.00 migration/0
    4 root      20   0     0    0    0 S  0.0  0.0   0:00.00 ksoftirqd/0
    5 root      RT   0     0    0    0 S  0.0  0.0   0:00.00 watchdog/0
    6 root      20   0     0    0    0 S  0.0  0.0   0:00.01 events/0
    7 root      20   0     0    0    0 S  0.0  0.0   0:00.00 cpuset
    8 root      20   0     0    0    0 S  0.0  0.0   0:00.00 khelper
    9 root      20   0     0    0    0 S  0.0  0.0   0:00.00 netns
   10 root      20   0     0    0    0 S  0.0  0.0   0:00.00 async/mgr
   11 root      20   0     0    0    0 S  0.0  0.0   0:00.00 pm
   12 root      20   0     0    0    0 S  0.0  0.0   0:00.00 sync_supers
   13 root      20   0     0    0    0 S  0.0  0.0   0:00.00 bdi-default
   14 root      20   0     0    0    0 S  0.0  0.0   0:00.00 kintegrityd/0
   15 root      20   0     0    0    0 S  0.0  0.0   0:00.01 kblockd/0
   16 root      20   0     0    0    0 S  0.0  0.0   0:00.00 kacpid
   17 root      20   0     0    0    0 S  0.0  0.0   0:00.00 kacpi_notify
   18 root      20   0     0    0    0 S  0.0  0.0   0:00.00 kacpi_hotplug
   19 root      20   0     0    0    0 S  0.0  0.0   0:00.41 ata/0
   20 root      20   0     0    0    0 S  0.0  0.0   0:00.00 ata_aux
   21 root      20   0     0    0    0 S  0.0  0.0   0:00.00 ksuspend_usbd
   22 root      20   0     0    0    0 S  0.0  0.0   0:00.00 khubd
```

No Items in Trash

root@localhost:~

To make on permanent basis, Create swap entry

```
[root@nehraclasses ~]# vim /etc/fstab
```

```
#
# /etc/fstab
# Created by anaconda on Wed Jul  2 19:40:42 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk'
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info
#
/dev/mapper/VolGroup-lv_root /                       ext4    defaults        1 1
UUID=14a2c075-ac25-4c24-80e4-76cca0b5cd13 /boot                   ext4    defaults        1 2
/dev/mapper/VolGroup-lv_swap swap                    swap    defaults        0 0
tmpfs                   /dev/shm                tmpfs   defaults        0 0
devpts                  /dev/pts                devpts  gid=5,mode=620  0 0
sysfs                   /sys                    sysfs   defaults        0 0
proc                    /proc                   proc    defaults        0 0
/swapfile swap swap defaults 0 0
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                                    16,34         All
root@localhost:~
```

Verify the entry

```
[root@localhost ~]# vim /etc/fstab
[root@localhost ~]# cat /etc/fstab | grep swap
/dev/mapper/VolGroup-lv_swap swap                    swap    defaults        0 0
/swapfile swap swap defaults 0 0
[root@localhost ~]#
No Items in Trash
root@localhost:~
```

Now check the all created swap

```
[root@localhost ~]# swapon -s
Filename                                Type            Size    Used    Priority
/dev/dm-1                               partition       4161528 0       -1
/swapfile                               file            1048568 0       -2
/swapfile2                              file            1048568 0       -3
[root@localhost ~]#
[root@localhost ~]# cat /proc/swaps
Filename                                Type            Size    Used    Priority
/dev/dm-1                               partition       4161528 0       -1
/swapfile                               file            1048568 0       -2
/swapfile2                              file            1048568 0       -3
[root@localhost ~]#
root@localhost:~
```